Ph.D. Student, Applied Mathematics
The Hong Kong Polytechnic University
25036713r@connect.polyu.hk | +86 189-4506-7854
djcdjcdjc2000.github.io

Junchao Duan

Research Focus

Current research develops fast numerical methods for differential stochastic linear complementarity problems, a class of nonsmooth stochastic dynamical systems. The work combines smoothing methods, sample average approximation, ultraspherical spectral discretization, and block iterative solvers, with an emphasis on structured error analysis and accuracy-cost evaluation.

Education

The Hong Kong Polytechnic University *2025.09 - 2028.06 (expected)*
Ph.D. in Applied Mathematics, Department of Applied Mathematics — Hong Kong SAR, China
Supervisor: Prof. Xiaojun Chen. Research direction: fast algorithms for differential stochastic linear complementarity problems.

Harbin Institute of Technology *2022.09 - 2025.06*
M.S. in Mathematics, School of Mathematics; rank: 1/18 — Harbin, China

Harbin Engineering University *2018.09 - 2022.06*
B.S. in Mathematics, School of Mathematical Sciences; rank: 3/96 — Harbin, China

Manuscripts

2026 — **Junchao Duan and Xiaojun Chen**. "An Ultraspherical Spectral Method for Differential Stochastic Linear Complementarity Problems." *Manuscript in preparation*.

Presentations

An Ultraspherical Spectral Method for Differential Stochastic Linear Complementarity Problems

May 2026	Poster	International Workshop on Numerical Methods and Scientific Computing, The Hong Kong Polytechnic University, Hong Kong SAR, China.
Apr. 2026	Oral	Special Session: Stochastic Optimization and Equilibrium, ICOTA 2026, Shanghai, China.
Mar. 2026	Poster	The 4th Guangzhou-Shenzhen Workshop on Nonsmooth Optimization, Guangzhou, China.

Teaching Experience

Teaching Assistant, Undergraduate Calculus *2025.09 - 2026.06*
Department of Applied Mathematics, The Hong Kong Polytechnic University — Hong Kong SAR, China

- Held weekly office hours for undergraduate Calculus students during Spring 2026, 4 hours per week.
- Assisted with homework grading and exam invigilation during Fall 2025 and Spring 2026.

Research Experience

Differential Stochastic Linear Complementarity Problems *2025 - present*
Ph.D. research, The Hong Kong Polytechnic University — Supervisor: Prof. Xiaojun Chen

- Developing a smoothing-SAA-spectral framework that combines Fischer-Burmeister smoothing, finite-sample stochastic feedback, ultraspherical spectral discretization, and block Gauss-Seidel solution of the coupled algebraic system.
- Establishing a layered error decomposition separating smoothing, sample approximation, spectral discretization, and iteration errors for computed trajectories.
- Designing numerical experiments and a data-driven Citi Bike station-inventory protocol to evaluate the accuracy-cost behavior of finite-sample spectral methods.

Sparse Spectral Methods for a Class of Optimal Control Problems *2024 - 2025*

M.S. thesis project, Harbin Institute of Technology

- Proposed a modal-basis pseudospectral scheme using Chebyshev polynomial approximations to transform constrained optimal control problems into sparse nonlinear programming problems.
- Awarded Outstanding Master's Thesis by Harbin Institute of Technology.

Selected Earlier Research Projects

Sparse subspace clustering	Reformulated an l_0-norm sparse representation model through an l_1-norm convex relaxation; implemented ADMM and soft-thresholding algorithms, obtaining clustering accuracy above 85% on YaleB and COIL-20 datasets.
Asteroid trajectory analysis	Built C++ routines for least-squares fitting, SVD-based trajectory estimation, and Ipopt-based spin-axis optimization, with numerical accuracy better than 10^{-6} in project experiments.

Selected Honors and Awards

2025	Outstanding Master's Thesis, Harbin Institute of Technology.
2025	Outstanding Graduate of Heilongjiang Province.
2024	National Scholarship for Graduate Students.
2022, 2023	First-Class Academic Scholarship.
2019-2021	National Scholarship (three times).
2019-2020	Selected mathematics and modeling competition awards, including a national third prize and provincial first/second prizes.

Research Skills

Methods	Numerical optimization, complementarity problems, stochastic approximation, spectral methods, nonlinear programming, scientific computing.
Programming	Python, C++, Matlab, Octave; experience with NumPy, Eigen, ADOL-C, and Ipopt.
Writing	LaTeX; academic writing and presentation in English.